Deriving the social cost of carbon

Chapter 6 of Hunt Allcott's *Environmental Economics Lecture Notes*

The **social cost of carbon (SCC)** is the welfare loss from one additional tonne of emissions, expressed in dollars of consumption today. Derive it by tracing a tonne into future consumption, then converting the resulting utility loss into today's dollars.

ϕ: damage slope e: emissions e': emissions response ρ: time preference r: consumption discount rate

R: discount factor η: utility curvature g: consumption growth f': capital return Teal annotations: consumption weight.

1. Start with resources and welfare

There are two periods and P identical people. Consumption c_t, income y_t, and spending K, M are per person. Spending K buys capital and M buys mitigation today. Capital pays $f(K)$ tomorrow; mitigation lowers per-person emissions $e(M)$, so $e'(M) < 0$. Total emissions are $Pe(M)$.

$$\begin{aligned} c_1 &= y_1 - K - M, \\ c_2 &= y_2 + f(K) + \underbrace{\phi P e(M)}_{\text{consumption loss from emissions}}, \qquad \phi < 0. \end{aligned}$$

One extra *total* tonne lowers each person's future consumption by $-\phi$ dollars, hence aggregate future damages are $(-\phi)P$ dollars per tonne. With pure time preference $\rho \geq 0$, the government chooses K, M to maximize

$$W = P\left[u(y_1 - K - M) + \frac{1}{1+\rho}\, u\big(y_2 + f(K) + \phi P e(M)\big)\right].$$

Assume positive consumption, increasing concave utility, and an interior optimum.

2. Differentiate mitigation spending, one link at a time

A small increase dM costs $P\,dM$ dollars today and avoids $P[-e'(M)]dM$ tonnes. The chain rule gives

$$\frac{1}{P}\frac{\partial W}{\partial M} = \underbrace{-u'(c_1)}_{\text{utility cost today}} + \frac{u'(c_2)}{1+\rho}\ \underbrace{\phi\ \underbrace{P e'(M)}_{\text{change in total emissions}}}_{\text{gain in consumption per person}} = 0.$$

Both ϕ and $e'(M)$ are negative, so the future gain is positive. Rearrange explicitly:

$$u'(c_1) = \frac{u'(c_2)}{1+\rho}\,\phi P e'(M) \quad \Longrightarrow \quad 1 = \frac{u'(c_2)}{(1+\rho)u'(c_1)}\,\phi P e'(M).$$

Divide by $-e'(M) > 0$ to obtain the notes' equation (6.11):

$$\underbrace{\frac{-1}{e'(M)}}_{\text{marginal abatement cost}} = \underbrace{(-\phi)P}_{\text{future dollars lost per tonne}}\ \underbrace{\frac{1}{1+\rho}\frac{u'(c_2)}{u'(c_1)}}_{\text{future dollars to today's dollars}} = \text{SCC}.$$

The left side is also $P\,dM/\{P[-e'(M)]dM\}$: dollars spent per tonne avoided. The right side converts damages into today's consumption units. In particular, dividing a utility loss by $u'(c_1)$ gives its dollar equivalent. **At the optimum, the last tonne avoided costs exactly the SCC.**

Global damages: whose consumption is lost?

3. Let emissions affect every country

Country i pays for mitigation. Country j receives its climate benefit. Each country has P_j identical people and per-person damage slope $\phi_j < 0$. Following Section 6.3, government i values everyone's utility equally, chooses only K_i, M_i, and cannot make transfers abroad. Other countries' investments are held fixed.

$$c_{j1} = y_{j1} - K_j - M_j,$$
$$c_{j2} = y_{j2} + f(K_j) + \phi_j \underbrace{\sum_k P_k e(M_k)}_{\text{total global emissions}}.$$

$$W = \sum_j P_j \left[u(c_{j1}) + \frac{u(c_{j2})}{1+\rho} \right].$$

Only country i loses consumption today; every country gains consumption tomorrow. For each j, the chain rule is

$$\frac{\partial c_{j2}}{\partial M_i} = \underbrace{\phi_j}_{\text{consumption change per tonne}} \underbrace{P_i e'(M_i)}_{\text{change in total emissions}}.$$

Insert this derivative into welfare and set the result to zero:

$$0 = \underbrace{-P_i u'(c_{i1})}_{\text{utility cost in country } i} + \frac{1}{1+\rho} \sum_j \underbrace{P_j u'(c_{j2}) \phi_j P_i e'(M_i)}_{\text{utility gain in country } j}.$$

Cancel P_i and divide by $u'(c_{i1})$, then by $-e'(M_i)$:

$$1 = \frac{e'(M_i)}{1+\rho} \sum_j P_j \phi_j \frac{u'(c_{j2})}{u'(c_{i1})},$$
$$\frac{-1}{e'(M_i)} = \frac{1}{1+\rho} \sum_j (-\phi_j) P_j \frac{u'(c_{j2})}{u'(c_{i1})}.$$

With isoelastic utility, $u(c) = c^{1-\eta}/(1-\eta)$ for $\eta \neq 1$, and $u(c) = \ln c$ for $\eta = 1$. Here $\eta \geq 0$ controls how quickly marginal utility falls. Both forms give $u'(c) = c^{-\eta}$, so

$$\frac{u'(c_{j2})}{u'(c_{i1})} = \frac{c_{j2}^{-\eta}}{c_{i1}^{-\eta}} = \left(\frac{c_{i1}}{c_{j2}} \right)^{\eta}.$$

Substituting yields the chapter's main global equation, (6.22):

$$\frac{-1}{e'(M_i)} = \mathrm{SCC}_i = \underbrace{\frac{1}{1+\rho}}_{\text{time preference}} \sum_j \underbrace{(-\phi_j) P_j}_{\text{future damages in } j} \underbrace{\left(\frac{c_{i1}}{c_{j2}} \right)^{\eta}}_{\text{marginal-utility weight}}.$$

4. Read the global SCC in consumption units

Each summand is **future dollars lost in j × their value in today's dollars in i**. The factor $1/(1+\rho)$ weights future utility; the ratio $u'(c_{j2})/u'(c_{i1})$ converts that utility into current consumption units. For $\eta > 0$, poorer future recipients or a richer country paying today raise the SCC, holding other terms fixed. With $\eta = 0$, all consumption weights equal one; with $\eta = 1$, someone consuming half as much has twice the marginal utility. The subscript i specifies whose current consumption values *global* damages.

What is being discounted? Deriving the Ramsey equation

5. Distinguish the rates from the factors

Pure time preference ρ discounts utility. One unit of future utility receives weight $1/(1+\rho)$ relative to one unit today, even if consumption is unchanged. This is a preference parameter.

The consumption discount rate r discounts dollars of consumption. A future dollar need not give the same utility as a current dollar. The rate r accounts for both time preference and this difference in marginal utility.

The discount factor R is the multiplier, not the rate:

$$R = \frac{1}{1+r}, \qquad \text{present value of future damages } D = RD.$$

A higher r means a smaller R and a lower SCC. For a constant annual rate and damages T years away, the factor is $(1+r)^{-T}$.

The capital return $f'(K)$ is a gross payoff. One dollar invested yields $f'(K)$ future dollars; $f'(K)-1$ is the net return. At the optimum it equals r.

6. Derive the consumption discount factor from welfare

In the single-country problem, a small future gain of d dollars per person offsets a current sacrifice of x dollars per person when

$$u'(c_1)x = \frac{u'(c_2)}{1+\rho}d \quad \Longrightarrow \quad R \equiv \frac{x}{d} = \underbrace{\frac{1}{1+\rho}}_{\text{discount future utility}} \underbrace{\frac{u'(c_2)}{u'(c_1)}}_{\text{convert utility to current dollars}}.$$

For capital, the first-order condition is $-u'(c_1) + u'(c_2)f'(K)/(1+\rho) = 0$. Hence

$$R = \frac{1}{f'(K)} = \frac{1}{1+r} \quad \Longrightarrow \quad r = f'(K) - 1.$$

The *descriptive* approach uses investment returns to value mitigation's opportunity cost. The *prescriptive* approach derives R from preferences and consumption. They coincide at this model's interior optimum.

7. Substitute consumption growth to get Ramsey

Let $g = c_2/c_1 - 1$ be *consumption growth*; η measures how fast marginal utility falls. Neither is a discount rate. Since $u'(c) = c^{-\eta}$,

$$\frac{1}{1+r} = R = \frac{1}{1+\rho}\left(\frac{c_1}{c_2}\right)^{\eta} = \frac{(1+g)^{-\eta}}{1+\rho},$$
$$1 + r = (1+\rho)(1+g)^{\eta},$$
$$\ln(1+r) = \ln(1+\rho) + \eta \ln(1+g).$$

Using $\ln(1+z) \approx z$ for small rates yields the Ramsey equation:

$$r \approx \underbrace{\rho}_{\text{pure time preference}} + \underbrace{\eta g}_{\text{consumption growth adjustment}}.$$

Exactly, $r = (1+\rho)(1+g)^{\eta} - 1$. If $g = 0$ or $\eta = 0$, then $r = \rho$; if $g > 0$ and $\eta > 0$, then $r > \rho$: future dollars give less utility because consumption is higher. All rates must use the same time interval.

Source: Chapter 6, pp. 34–40; especially equations (6.3)–(6.11) and (6.17)–(6.22). This derivation retains the chapter's two-period, linear-damage model.